

Myocardial Perfusion Imaging, Pharmacologic Stress

ACG: A-0079 (AC)

[Link to Codes](#)

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Clinical Indications

- Myocardial perfusion imaging, pharmacologic stress, may be indicated for **1 or more** of the following(1):
 - ☐ Acute chest pain or acute anginal equivalent (eg, jaw pain, neck pain, nausea and vomiting)[A] in patient presenting to emergency department or chest pain center and **ALL** of the following(13)(25)(28)(29)(30)(31)(32):[N](#)
 - Cardiac risk factors present, as indicated by **1 or more** of the following(35)(36)(37):
 - Age 45 years or older for male patient
 - Age 55 years or older for female patient, or postmenopausal(38)
 - Autoimmune disease (eg, lupus, psoriasis, rheumatoid arthritis)(39)
 - Cancer survivor previously treated with potentially cardiotoxic therapy (eg, chest radiation, fluoropyrimidines)(40)(41)(42)
 - Chronic kidney disease
 - Diabetes mellitus
 - Dyslipidemia
 - Erectile dysfunction
 - Family history of premature atherosclerotic cardiovascular disease[B] (ie, female younger than 65 years, male younger than 55 years)
 - History of heart failure
 - HIV infection
 - Hypertension
 - Metabolic syndrome
 - Obesity
 - Peripheral arterial disease or prior cerebrovascular disease
 - Premature menopause (ie, age younger than 40 years)
 - Previous myocardial infarction
 - Previous revascularization (ie, percutaneous coronary intervention or coronary artery bypass grafting) or medical treatment of coronary artery disease(13)
 - Sedentary lifestyle
 - Sleep disorder (eg, obstructive sleep apnea)
 - South Asian ancestry(43)(44)

- Tobacco use
- Woman with history of eclampsia, preeclampsia, or preterm delivery
- ECG nondiagnostic
- Negative or minimally elevated cardiac biomarkers
- Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)

☐ Asymptomatic patient and **ALL** of the following(25):[\[1\]](#)

- Coronary heart disease, high-risk, as indicated by **1 or more** of the following(36)(48):
 - Calculated 10-year absolute coronary heart disease risk greater than 20% by global risk calculator(35)(49)
 - Coronary heart disease risk equivalent (eg, peripheral arterial disease, carotid artery disease, or abdominal aortic aneurysm)
 - Diabetes mellitus in patient age 40 years or older
 - Prior stroke or transient ischemic attack(50)
- Exercise treadmill testing alone is or would be unreliable or inconclusive due to **1 or more** of the following(1)(25)(30)(38):
 - Anginal symptoms induced by exercise in high-risk patient without any diagnostic ECG changes on previous exercise treadmill test(51)
 - Atrial fibrillation or flutter
 - Electronically paced ventricular rhythm
 - Insufficient workload (ie, inability to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(8)
 - Left bundle branch block(52)
 - Left ventricular hypertrophy with repolarization abnormalities on baseline ECG
 - Patient taking digoxin
 - Pre-excitation syndrome (eg, Wolff-Parkinson-White syndrome)
 - ST segment on baseline ECG with 1 mm depression or greater
- Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)

☐ Coronary artery disease, known, and need for further evaluation, as indicated by **ALL** of the following(10)(25)(26)(53)(54)(55)(56):[\[1\]](#)

- Clinically significant coronary artery disease, as indicated by **1 or more** of the following:
 - Anomalous coronary artery, known, and need for functional assessment(47)(51)(62)(63)(64)
 - Before revascularization (ie, percutaneous coronary intervention or coronary artery bypass grafting) to demonstrate ischemia
 - Cardiac CT angiography with 50% or greater stenosis of at least one coronary artery, or inconclusive results
 - Change in clinical condition, as indicated by **1 or more** of the following(13)(30):
 - Increase in ischemic symptom frequency or severity
 - Increase of heart failure symptoms
 - New onset of chest pain or atypical chest pain without ECG changes
 - New ST-segment changes on ECG at rest
 - Ongoing chest pain despite optimal medical therapy
 - Coronary calcium Agatston score 1 to 99 on cardiac CT scan, and high risk for coronary heart disease, as indicated by **1 or more** of the following(26)(36)(60)(61)(65):
 - Calculated 10-year absolute coronary heart disease risk 20% or greater by global risk calculator(35)(49)
 - Coronary heart disease risk equivalent (eg, peripheral arterial disease, carotid artery disease, or abdominal aortic aneurysm)
 - Diabetes mellitus in patient age 40 years or older
 - Prior stroke or transient ischemic attack(50)
 - Coronary calcium Agatston score of 100 or more on cardiac CT scan(60)(61)(65)
 - Detection of recurrent stenosis or progression of disease in asymptomatic patient who had silent ischemia prior to revascularization and **1 or more** of the following(25)(26)(27):
 - Diabetes
 - Five years or more after coronary artery bypass grafting

- History of prior cardiac arrest
 - Incomplete revascularization procedure (ie, persistent stenosis after percutaneous coronary intervention or coronary artery bypass grafting)
 - Involvement of proximal left anterior descending artery
 - Left ventricular ejection fraction of 35% or less
 - Two years or more after percutaneous coronary intervention (ie, angioplasty or stent)
- Following acute coronary syndrome and **ALL** of the following(25)(66):
 - Absence of current ischemic symptoms
 - No signs of heart failure or hemodynamic instability(13)
 - Not planning to undergo early revascularization
- Following ST-segment elevation myocardial infarction
- History of Kawasaki disease with known coronary artery aneurysm, and need to demonstrate inducible ischemia(47)(67)
- Stable ischemic heart disease and **ALL** of the following(53)(68):
 - New, recurrent, or worsening symptoms
 - Symptoms are not consistent with unstable angina.
- Exercise treadmill testing alone is or would be unreliable or inconclusive due to **1 or more** of the following(1)(25)(30)(38):
 - Anginal symptoms induced by exercise in high-risk patient without any diagnostic ECG changes on previous exercise treadmill test(51)
 - Atrial fibrillation or flutter
 - Electronically paced ventricular rhythm
 - Insufficient workload (ie, inability to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(8)
 - Left bundle branch block(52)
 - Left ventricular hypertrophy with repolarization abnormalities on baseline ECG
 - Patient taking digoxin
 - Pre-excitation syndrome (eg, Wolff-Parkinson-White syndrome)
 - ST segment on baseline ECG with 1 mm depression or greater
- Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)

☐ Coronary artery disease, suspected, as indicated by **ALL** of the following(25)(30)(31)(55)(56):**N**

- Clinical condition or test results, as indicated by **1 or more** of the following:
 - Arrhythmia, as indicated by **1 or more** of the following:
 - Frequent premature ventricular contractions (one or more on 12-lead electrocardiogram or more than 30 per hour on ambulatory Holter monitoring), new onset(74)
 - Prior to initiation of class IC antiarrhythmic medication (eg, flecainide)(75)(76)
 - Supraventricular tachycardia, new onset, in patient with cardiac risk factors, as indicated by **1 or more** of the following(35)(36)(48)(77):
 - Age 45 years or older for male patient
 - Age 55 years or older for female patient, or postmenopausal(38)
 - Autoimmune disease (eg, lupus, psoriasis, rheumatoid arthritis)
 - Cancer survivor previously treated with potentially cardiotoxic therapy (eg, chest radiation, fluoropyrimidines)(40)(41)(42)
 - Chronic kidney disease
 - Diabetes mellitus
 - Dyslipidemia
 - Erectile dysfunction
 - Family history of premature atherosclerotic cardiovascular disease^[B] (ie, female younger than 65 years, male younger than 55 years)
 - History of heart failure
 - HIV infection
 - Hypertension
 - Metabolic syndrome
 - Obesity(78)
 - Peripheral arterial disease or prior cerebrovascular disease
 - Premature menopause (ie, age younger than 40 years)
 - Previous myocardial infarction

- Previous revascularization (ie, percutaneous coronary intervention or coronary artery bypass grafting) or medical treatment of coronary artery disease
 - Sedentary lifestyle(79)
 - Sleep disorder (eg, obstructive sleep apnea)
 - South Asian ancestry(43)
 - Tobacco use
 - Woman with history of eclampsia, preeclampsia, or preterm delivery
 - Ventricular tachycardia
 - Asymptomatic patient with elevated troponin level(80)
 - Asymptomatic patient with ischemic changes present on resting ECG
 - Chronic secondary mitral regurgitation, to establish possible ischemic etiology(81)(82)
 - Congenital heart disease, known (eg, repaired aortic coarctation, previous coronary reimplantation surgery, coronary artery compression)(47)(72)(73)(83)
 - Equivocal results on exercise treadmill test or cardiac CT angiography that require further evaluation(13)
 - Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)
- ☐ Diabetes mellitus, as indicated by **ALL** of the following(84)(85):[N](#)
- High clinical suspicion of coronary artery disease, as indicated by **1 or more** of the following:
 - Abnormal resting ECG
 - Angina, atypical chest pain, or anginal equivalent (eg, exertional dyspnea)^[A](87)
 - Autonomic neuropathy
 - Peripheral arterial disease
 - Signs or symptoms of vascular disease (eg, carotid bruit, claudication)
 - Stroke or transient ischemic attack
 - Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)
- ☐ Heart failure, cardiomyopathy, or ventricular dysfunction, known or suspected, and **ALL** of the following(88)(89)(90):[N](#)
- Need for additional evaluation, as indicated by **1 or more** of the following(8)(47)(88)(92)(93)(94):
 - Assessment for myocardial viability to determine need for revascularization
 - Diagnosis of cardiomyopathy in patient with systemic disease (eg, sarcoidosis, amyloidosis)(95)(96)
 - Evaluation of left ventricular function and ejection fraction prior to receiving doxorubicin or other cardiotoxic drug for cancer chemotherapy(97)(98)(99)
 - Evaluation of ventricular dysfunction when echocardiogram is nondiagnostic(100)(101)
 - Need to differentiate ischemic from nonischemic etiology in unexplained dilated cardiomyopathy
 - Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)
- ☐ Nonacute chest pain or ischemic equivalent, as indicated by **ALL** of the following(13)(25)(31)(56):[N](#)
- Cardiac risk factors present, as indicated by **1 or more** of the following(35)(36)(37):
 - Age 45 years or older for male patient
 - Age 55 years or older for female patient, or postmenopausal(38)
 - Autoimmune disease (eg, lupus, psoriasis, rheumatoid arthritis)(39)
 - Cancer survivor previously treated with potentially cardiotoxic therapy (eg, chest radiation, fluoropyrimidines)(40)(41)(42)
 - Chronic kidney disease
 - Diabetes mellitus
 - Dyslipidemia
 - Erectile dysfunction

- Family history of premature atherosclerotic cardiovascular disease^[B] (ie, female younger than 65 years, male younger than 55 years)
- History of heart failure
- HIV infection
- Hypertension
- Metabolic syndrome
- Obesity
- Peripheral arterial disease or prior cerebrovascular disease
- Premature menopause (ie, age younger than 40 years)
- Previous myocardial infarction
- Previous revascularization (ie, percutaneous coronary intervention or coronary artery bypass grafting) or medical treatment of coronary artery disease⁽¹³⁾
- Sedentary lifestyle
- Sleep disorder (eg, obstructive sleep apnea)
- South Asian ancestry⁽⁴³⁾⁽⁴⁴⁾
- Tobacco use
- Woman with history of eclampsia, preeclampsia, or preterm delivery
- Evaluation of chest pain needed, as indicated by **1 or more** of the following⁽¹³⁾⁽²⁶⁾:
 - Likely cardiac pain, as indicated by **ALL** of the following⁽¹⁰³⁾:
 - Deep, poorly localized chest, arm, jaw, neck, shoulder, or upper abdominal pain or discomfort
 - Relieved within 5 to 15 minutes by rest or sublingual nitroglycerin
 - Reproducibly associated with physical exertion or emotional stress
 - Possible cardiac pain (ie, symptom not classically cardiac in nature) and **1 or more** of the following:
 - Male age 40 years or older
 - Female age 50 years or older or postmenopausal^[C]⁽¹⁰⁵⁾
 - Coronary artery calcium score of 100 or more⁽¹⁰⁶⁾
- Exercise treadmill testing alone is or would be unreliable or inconclusive due to **1 or more** of the following⁽¹⁾⁽²⁵⁾⁽³⁰⁾⁽³⁸⁾:
 - Anginal symptoms induced by exercise in high-risk patient without any diagnostic ECG changes on previous exercise treadmill test⁽⁵¹⁾
 - Atrial fibrillation or flutter
 - Electronically paced ventricular rhythm
 - Insufficient workload (ie, inability to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)⁽⁸⁾
 - Left bundle branch block⁽⁵²⁾
 - Left ventricular hypertrophy with repolarization abnormalities on baseline ECG
 - Patient taking digoxin
 - Pre-excitation syndrome (eg, Wolff-Parkinson-White syndrome)
 - ST segment on baseline ECG with 1 mm depression or greater
- Need for pharmacologic testing, as indicated by **1 or more** of the following⁽⁸⁾⁽²⁶⁾⁽⁴⁵⁾⁽⁴⁶⁾:
 - Electronically paced rhythm⁽³⁰⁾
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG⁽³⁰⁾
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)⁽⁴⁷⁾

☐ Preoperative or preprocedural cardiovascular evaluation needed, as indicated by **1 or more** of the following⁽⁹⁰⁾⁽¹⁰⁷⁾:^[N]

- Patient undergoing solid organ transplant
- Prior to arterial vascular surgery (eg, aortic aneurysm, aortoiliac or aortofemoral bypass) in patient with poor or unknown functional capacity (eg, not able to climb 2 flights of stairs or walk up hill), as indicated by **ALL** of the following⁽¹⁰⁷⁾:
 - Clinical risk factors, as indicated by **1 or more** of the following:
 - Chronic kidney disease
 - Diabetes mellitus
 - History of arrhythmia
 - History of cerebrovascular disease
 - History of heart disease (eg, ischemic, valvular)
 - History of heart failure
 - Peripheral arterial disease (eg, carotid artery disease, abdominal aortic aneurysm, lower extremity arterial disease)
 - Exercise treadmill testing is or would be unreliable or inconclusive due to **1 or more** of the following⁽¹⁾:
 - Anginal symptoms induced by exercise in high-risk patient without any diagnostic ECG changes on previous exercise treadmill test⁽⁵¹⁾
 - Atrial fibrillation or flutter

- Electronically paced ventricular rhythm
 - Insufficient workload (ie, inability to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)
 - Left bundle branch block
 - Left ventricular hypertrophy with repolarization abnormalities on baseline ECG
 - Patient taking digoxin
 - Pre-excitation syndrome (eg, Wolff-Parkinson-White syndrome)
 - ST segment on baseline ECG with 1 mm depression or greater
- Need for pharmacologic testing, as indicated by **1 or more** of the following:
 - Electronically paced rhythm or left bundle branch block
 - Exercise contraindicated (eg, severe hypertension or severe aortic stenosis)
 - Left ventricular hypertrophy with repolarization abnormalities on baseline ECG
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)
 - Pre-excitation syndrome (eg, Wolff-Parkinson-White syndrome)
- Prior to intermediate-risk or high-risk noncardiac surgery (eg, abdominal or thoracic surgery, hip replacement) in patient with poor or unknown functional capacity (eg, not able to climb 2 flights of stairs or walk up hill) and **ALL** of the following:
 - Clinical risk factors, as indicated by **1 or more** of the following(107)(109):
 - Chronic kidney disease
 - Diabetes mellitus
 - History of arrhythmia
 - History of cerebrovascular disease
 - History of heart disease (eg, ischemic, valvular)
 - History of heart failure
 - Peripheral arterial disease (eg, carotid artery disease, abdominal aortic aneurysm, lower extremity arterial disease)
 - Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)

☐ Syncope and **ALL** of the following(25):[\[B\]](#)

- Cardiac risk factors present, as indicated by **1 or more** of the following(35)(36)(37):
 - Age 45 years or older for male patient
 - Age 55 years or older for female patient, or postmenopausal(38)
 - Autoimmune disease (eg, lupus, psoriasis, rheumatoid arthritis)(39)
 - Cancer survivor previously treated with potentially cardiotoxic therapy (eg, chest radiation, fluoropyrimidines)(40)(41)(42)
 - Chronic kidney disease
 - Diabetes mellitus
 - Dyslipidemia
 - Erectile dysfunction
 - Family history of premature atherosclerotic cardiovascular disease[B] (ie, female younger than 65 years, male younger than 55 years)
 - History of heart failure
 - HIV infection
 - Hypertension
 - Metabolic syndrome
 - Obesity
 - Peripheral arterial disease or prior cerebrovascular disease
 - Premature menopause (ie, age younger than 40 years)
 - Previous myocardial infarction
 - Previous revascularization (ie, percutaneous coronary intervention or coronary artery bypass grafting) or medical treatment of coronary artery disease(13)
 - Sedentary lifestyle
 - Sleep disorder (eg, obstructive sleep apnea)
 - South Asian ancestry(43)(44)
 - Tobacco use
 - Woman with history of eclampsia, preeclampsia, or preterm delivery
- Exercise treadmill testing alone is or would be unreliable or inconclusive due to **1 or more** of the following(1)(25)(30)(38):

- Anginal symptoms induced by exercise in high-risk patient without any diagnostic ECG changes on previous exercise treadmill test(51)
- Atrial fibrillation or flutter
- Electronically paced ventricular rhythm
- Insufficient workload (ie, inability to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(8)
- Left bundle branch block(52)
- Left ventricular hypertrophy with repolarization abnormalities on baseline ECG
- Patient taking digoxin
- Pre-excitation syndrome (eg, Wolff-Parkinson-White syndrome)
- ST segment on baseline ECG with 1 mm depression or greater
- Need for pharmacologic testing, as indicated by **1 or more** of the following(8)(26)(45)(46):
 - Electronically paced rhythm(30)
 - Exercise contraindicated (eg, uncontrolled hypertension or severe aortic disease such as aortic coarctation or thoracic aortic aneurysm or dissection)
 - Left bundle branch block on ECG(30)
 - Patient unable to exercise or to achieve sufficient workload (ie, unable to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(47)
- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

Pharmacologic stress myocardial perfusion imaging, such as pharmacologic stress SPECT, involves intravenous injection of a radioactive tracer (eg, thallium, sestamibi, or tetrofosmin), which is taken up by myocardial cells and visualized by a digital gamma camera, thereby reflecting the distribution of blood perfusion throughout the myocardium.(1)(2)(3) **(EG 2)** Coronary hyperemia is induced by a vasodilator, such as adenosine, dipyridamole, or regadenoson,(4)(5)(6) or an adrenergic agent such as dobutamine, in lieu of stress via exercise or in addition to submaximal exercise.(1)(3)(7)(8) **(EG 2)** A defect in the image with stress that is not present at rest usually indicates an area of myocardial ischemia.(3) **(EG 2)** Myocardial perfusion imaging synchronized with ECG (eg, gated SPECT) can assess ventricular function, including ejection fraction, in addition to myocardial perfusion.(2)(8) **(EG 2)** Pharmacologic stress testing using the vasodilator agents adenosine and dipyridamole is contraindicated in patients with severe reactive airway disease (eg, asthma or chronic obstructive pulmonary disease) because of provocation of bronchospasm; regadenoson or dobutamine may be substituted in this population.(8)(9)(10) **(EG 2)**

Myocardial perfusion imaging has been noted by specialty societies to have the most clinical utility in patients who are at intermediate risk for coronary artery disease, in those requiring management or prognostic information, and in those with unexplained and persistent symptoms.(11)(12)(13) **(EG 2)** Myocardial perfusion imaging systems that combine SPECT and CT technology (also known as "hybrid" systems) are now widely available.(8)(14)(15)(16) **(EG 2)** It has been noted that myocardial perfusion scans contribute at least 20% of the estimated annual collective radiation dose in the United States, although the lifetime cancer risk from a single myocardial perfusion imaging study is thought to be small.(17)(18) **(EG 2)** Best-practice methods to maximize diagnostic quality while minimizing radiation exposure have been proposed.(8)(19)(20)(21) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 68 published peer reviewed articles, 38 specialty society or other evidence-based guidelines, and 1 book section.

For acute chest pain or acute anginal equivalent, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Myocardial perfusion stress and rest imaging is useful for diagnosing ischemia in low-risk to intermediate-risk patients who have a nondiagnostic ECG and negative cardiac biomarkers.(25)(26)(33) **(EG 2)** Myocardial perfusion imaging is useful in patients with chest pain for whom exercise treadmill testing would be unreliable, such as those with complete left bundle branch block or an electronically paced rhythm.(26) **(EG 2)** A meta-analysis determined that myocardial perfusion imaging and stress echocardiography have similar high negative predictive values for myocardial infarction and cardiac death; the meta-analysis concluded that by accurately identifying patients as low risk, additional interventions such as cardiac catheterization can be avoided.(34) **(EG 1)** A systematic review and meta-analysis of 77 studies involving a total of 49,541 patients presenting to the emergency department with chest pain concluded that myocardial perfusion scintigraphy had a sensitivity of 85% and specificity of 92% for the detection of acute coronary syndrome.(29) **(EG 1)**

For asymptomatic patients, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An appropriate-use criteria statement endorsed by multiple specialty societies notes that pharmacologic myocardial perfusion stress imaging is appropriate in asymptomatic patients who are at high risk for coronary heart disease.(25) **(EG 2)**

For coronary artery disease (known), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Myocardial perfusion imaging is useful for detecting the presence of new or worsening ischemia suggestive of myocardium-at-risk and identifying the need for changes to treatment plans, including revascularization; it is also used to detect restenosis or new areas of stenosis following revascularization.(10)(25)(33)(53) **(EG 2)** A meta-analysis of 166 studies found that, among myocardial perfusion imaging techniques, stress SPECT is widely available and the most extensively validated.(57) **(EG 2)** A prospective study of 266 patients with known coronary artery disease found that a normal myocardial perfusion imaging scan was predictive of a favorable long-term prognosis.(58) **(EG 2)** Pharmacologic myocardial perfusion imaging is useful for localizing cardiac ischemia in patients who are unable to exercise or cannot achieve target heart rates (usually 85% of predicted maximum heart rate on a standard Bruce protocol) with exercise stress testing, and in patients with an ECG that is not interpretable for ischemia, such as those with a complete left bundle branch block.(25) **(EG 2)** A study of 1126 generally asymptomatic patients without previous cardiovascular disease who had both a cardiac CT scan with a standard Agatston criteria score and a myocardial perfusion stress imaging scan by SPECT showed that 31% of patients with a score greater than 400 had an abnormal SPECT result.(59) **(EG 2)** Review articles and specialty society guidelines note that a coronary artery calcium score of 100 or more Agatston units indicates a high plaque burden with moderately to severely increased cardiovascular risk.(13)(60)(61) **(EG 2)** A review article and an appropriate-use criteria document note that myocardial perfusion stress imaging is appropriate to detect ischemia and evaluate function in patients with an anomalous aortic origin of a coronary artery.(51)(62) **(EG 2)**

For coronary artery disease (suspected), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Appropriate-use criteria statements endorsed by multiple specialty societies note that pharmacologic myocardial perfusion stress imaging is appropriate for evaluation of symptomatic patients with intermediate pretest probability who are suspected of having coronary artery disease.(25)(26) **(EG 2)** A systematic comparative effectiveness review and meta-analysis of 30 studies found that myocardial perfusion stress imaging has a summary sensitivity of 81% and specificity of 78% for the detection of coronary artery disease in women.(69) **(EG 2)** A retrospective study of 473 patients with limited exercise capacity and suspected coronary artery disease found that dobutamine stress SPECT imaging provided incremental prognostic value in addition to clinical and stress test parameters.(70) **(EG 2)** A specialty society consensus statement and a review article note that alternative diagnostic testing without radiation exposure should be considered for premenopausal women with suspected ischemic heart disease.(38)(71) **(EG 2)** A review article notes that certain congenital heart conditions are associated with risk factors for ischemic heart disease (eg, persistent hypertension in patients with repaired aortic coarctation); coronary artery compression syndromes and coronary fistulas can cause coronary ischemia, as can congenital heart surgeries that involve coronary reimplantation. Pharmacologic myocardial perfusion stress imaging can detect ischemic heart disease in these patients.(72) **(EG 2)** Specialty society guidelines recommend periodic noninvasive ischemia provocation testing, which may include pharmacologic myocardial perfusion stress imaging in combination with echocardiography, for adult patients who have undergone an arterial switch repair of transposition of the great vessels. These guidelines also note that stress imaging can be useful for the evaluation of myocardial ischemia in adult patients with unrepaired congenital heart disease.(62)(73) **(EG 2)**

For diabetes mellitus, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** There is no demonstrated clinical benefit to routine screening of asymptomatic diabetic patients, but candidates for cardiac testing include those with typical or atypical cardiac symptoms, signs or symptoms of vascular disease, or an abnormal resting ECG.(84)(86) **(EG 2)** Appropriate-use criteria endorsed by multiple specialty societies note that the presence of diabetes contributes to a higher pretest likelihood of coronary artery disease, making the use of noninvasive imaging such as pharmacologic myocardial perfusion stress imaging reasonable as an initial diagnostic test in patients who are suspected to have ischemic heart disease.(26) **(EG 2)** A study of 319 type 2 diabetic patients with stable mild anginal symptoms who underwent myocardial perfusion stress imaging by SPECT showed that those with severe demonstrated myocardial ischemia are at 3 to 6 times higher risk of cardiac death and nonfatal myocardial infarction than those with no or moderate myocardial ischemia, suggesting a more aggressive and invasive approach for those with severe ischemia.(87) **(EG 2)**

For heart failure, cardiomyopathy, or ventricular dysfunction, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Myocardial perfusion stress imaging is more sensitive but less specific than dobutamine stress echocardiography in predicting ventricular function improvement after revascularization (eg, percutaneous coronary intervention).(91) **(EG 2)** Specialty society guidelines state that myocardial perfusion stress imaging may be useful in assessing ischemia and myocardial viability in patients with heart failure.(92)(93) **(EG 2)**

For nonacute chest pain or ischemic equivalent, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Appropriate-use criteria statements endorsed by multiple specialty societies note that pharmacologic myocardial perfusion stress imaging is appropriate for evaluation of symptomatic patients with intermediate pretest probability who are suspected of having coronary artery disease.(25)(26) **(EG 2)** A retrospective cohort study of 1,491,642 patients with nonacute chest pain compared outcomes between patients who underwent noninvasive diagnostic testing (exercise treadmill test, myocardial perfusion scan, stress echocardiography, coronary CT scan) with patients who did not have diagnostic testing and found, at a median follow-up of 4.1 years, that patients who underwent noninvasive testing had a 25% lower rate of a composite outcome of cardiovascular mortality, acute myocardial infarction, or unstable angina compared with patients who had no cardiac testing.(102) **(EG 2)**

For preoperative cardiovascular evaluation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Myocardial perfusion imaging may be indicated for those circumstances in which the test results will clearly affect management of high-risk patients (eg, cardiovascular or cerebrovascular disease, diabetes, chronic kidney disease) prior to either cardiovascular surgery or noncardiac intermediate-risk surgeries; however, myocardial perfusion has not supplanted coronary angiography in the preoperative evaluation of possible ischemia in patients prior to surgery for aortic stenosis.(107)(108) **(EG 2)**

For syncope, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An appropriate-use criteria statement endorsed by multiple specialty societies notes that pharmacologic myocardial perfusion stress imaging is appropriate for the evaluation of syncope in patients who are at intermediate or high risk for coronary heart disease.(25) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For follow-up of normal stress SPECT studies, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A study investigating the timing and utility of routine follow-up stress SPECT scans found that these follow-up scans have minimal impact on referral to revascularization.(22) **(EG 2)**

For routine screening for coronary artery disease, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** Specialty society guidelines note that the use of SPECT imaging is inappropriate for asymptomatic patients at low risk for coronary artery disease.(23)(24)(25)(26) **(EG 2)** In addition, for all patients, a specialty society statement recommends that serial myocardial perfusion imaging testing be avoided.(11) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(110); 42 CFR 422.101(111)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(112); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(113); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(114)

Medicare Coverage Determinations: Medicare Coverage Database(115)

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Footnotes

[A] Symptoms of chest pain or pressure, chest tightness, burning, dyspnea, diaphoresis, shoulder pain, palpitations, syncope, light-headedness, intense fatigue, nausea or vomiting, and jaw pain may represent angina or an anginal equivalent (ie, a symptom due to ischemic heart disease).(13)(25)(27) [A in Context Link 1, 2, 3]

[B] Atherosclerotic cardiovascular disease (ASCVD) includes acute coronary syndrome, coronary or other arterial revascularization, history of myocardial infarction, peripheral arterial disease (including aortic aneurysm) of atherosclerotic origin, stable or unstable angina, stroke, or transient ischemic attack.(36) [B in Context Link 1, 2, 3, 4]

[C] Women with coronary artery disease have more accompanying symptoms, such as dyspnea, nausea, and fatigue, than do men. They may also present with prodromal symptoms.(13)(104) [C in Context Link 1]

Codes

CPT®: 0742T, 78451, 78452, 78453, 78454

HCPCS: A9500, A9502, A9505, J0153, J1245, J1250, J2785

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